INCH-POUND

A-A-54841
31 March 1993

COMMERCIAL ITEM DESCRIPTION

SAW, BONE, ELECTRIC, OSCILLATING, AUTOPSY

The General Services Administration has authorized the use of this Commercial Item Description as a replacement for Military Specification MIL-S-36703(2) which is cancelled.

This Commercial Item Description covers an electric motor driven saw, with high speed oscillating drive, suitable for cutting human bone in autopsy techniques.

Salient characteristics:

Style, design, and dimensions. Shall be an electric motor driven autopsy saw, suitable for cutting human bone in autopsy procedures without excessively damaging soft tissue. Saw shall be of the non-rotating, high frequency, oscillating design, complete with an oscillating head, two blades, one blade arbor assembly, one Allen wrench, one arbor nut wrench, and spare parts as specified herein. Style, design and dimensions shall be in accordance with Figures 1 and 2. Shall conform to the applicable requirements of Underwriters' Laboratories Inc. Standard UL 544 or the latest UL Standard for Medical and Dental Equipment.

Motor and switch. Motor shall operate on either 115 volt direct current or 115 volt, single phase, 60 cycles per second, alternating current. Motor shall be fan cooled, permanently lubricated, universal type, rated at 0.1 horsepower at 10,000 revolutions per minute (RPM) continuous. Brush holders shall be protected by durable, corrosion-resistant plugs threaded into and flush with the saw housing. Saw shall have "off" and "on" switch (properly identified) mounted on motor. Simple and effective means shall be provided for removing and replacing motor brushes.

AMSC N/A FSC 6515

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Power cable and plug. The saw shall be furnished with a 3-wire, electrical power cord, 9 feet minimum length, and a 3-prong plug in accordance with ANSI Standard C72.11. The power cable shall be provided with either a spring type or an integrally molded component to assure strain relief of power cable at the housing end.

Blade arbor shaft assembly. The centerline of the blade arbor shall be offset from the centerline of the motor shaft, as shown on drawing, to provide blade in a safe and convenient position for saw operation. The arbor shaft assembly shall be supported in sleeve bearings on each side of open link mechanism that transmits rotating motion of motor into oscillatory motion of the blade. Blade arbor shall be designed to provide a proper and secure fit for the blades specified herein. As an alternate, the blade arbor assembly may be supplied permanently attached to the blade, the arbor nut wrench need not be supplied.

Center of balance. With supply cable supported or detached, the center of balance of the saw assembly shall be 6-1/2 ± 1/2 inches from the front side of the arbor shaft assembly (measured parallel to the longitudinal axis of the power handle) as shown on drawing.

Drive mechanism. The rotary power of the motor shall be transmitted into an oscillatory action, at the blade end, by means of a cam and open link mechanism. The cam shall be equipped with a roller bearing. The open link mechanism shall be mechanically fastened to the offset blade arbor to suitably transmit the required oscillation rate. Gear transmission design shall not be acceptable.

Housing The working mechanism shall be enclosed by a compact, rigid and thoroughly protected corrosion-resistant housing. Housing shall provide intake and outlet ports for oscillating mechanism. The housing, except for the ventilating openings, shall totally enclose all stationary and rotating electrical parts. All external fastening devices shall be recessed into the housing. Contours of mating housing parts shall match, with no sharp corners or edges. Housing components shall be accurately fitted with adequate oil seals and gaskets to prevent oil leakage. The distal end of the housing shall be fitted with a shaft extension (chuck) to permit secure fit of the arbor. The chuck shall be provided with an Allen head setscrew for securing the blade arbor within the chuck.

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Blade. Each autopsy saw shall be furnished with two (2) large-sectional blades of the Stryker design and conform to Figure 2. The blades shall be suitable for cutting human bone and shall have 22 teeth per inch, 60 "V" type, set 0.005 ± 0.002 inch each side. The hardness shall be, after proper stress releaving, Rockwell C-40 minimum for type 410 material and C-49 minimum for type 420 material. Finish shall be 4 micro inches roughness height in accordance with ANSI Standard B46.

Saw weight. The autopsy saw, including power cable, 3-prong plug and blade, shall not exceed 5 pounds (less carrying case, wrench and remaining blade).

Fastening devices. All screws, pins, bolts and similar parts shall be installed with means of preventing loss of tightness.

Wrenches. Each autopsy saw shall be furnished with an Allen wrench, as shown in Figure 1, for use in changing blades to arbor of the saw. An arbor nut wrench shall be furnished to facilitate change of blades on the blade arbor. Wrenches shall be fabricated from corrosion-resistant metal. If the arbor assembly is supplied permanently attached to the blade, the arbor nut wrench need not be supplied.

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Performance. With input voltage set at 115 volts, the unit shall be capable of meeting the following performance characteristics:

Characteristic	Requirement
No load cutting strokes (oscillations) per minute (Free running)	17,000 minimum
* Full load oscillations under full load current of 1.2 amperes	13,000 minimum
No load current (Free running)	0.7 amperes maximum
* Temperature rise of motor housing during 5 minute operating period, under full load current of 1.2 amperes (when operated from a cold start at 75° to 76° Fahrenheit)	40 Degrees Fahrenheit maximum over ambient temperature
* Continuous operation without breakdown or failure, under full load current of 1.2 amperes without loss to cutting ability	100 hours minimum

* Full load current shall be defined as occurring when the output shaft is loaded with a mass of 7.0 ± 1.0 ounces (reference steel mass dimensions of 2 inches outer diameter and a height of 1/2 inch). The full load current of 1 2 amperes shall be read on a suitable ammeter at the loaded condition.

NOTE: This test may be waived upon request by the contractor, when the contractor has previously furnished items in accordance with the requirements of this specification and performed this test, and there has been no major design change of motor and drive mechanism

Carrying case. The autopsy saw shall be provided with a durable aluminum carrying case suitably fitted with cushioning material which shall secure all components during transit. In addition, the case shall be compartmentalized to secure all components.

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Carrying case identification. A metal plate shall be secured to the top surface of the carrying case which shall be permanently stamped or etched with the letters "U.S.", the National stock number, and the contractor's name or registered trademark.

Spare parts. In addition to the components listed above, the autopsy saw shall be supplied with the following parts which shall be secured within the carrying case:

Spare Part	Quantity
Arbor	1
Slotted Collar	1
Nut	1
Allen wrench	2
Allen head setscrew	4

The arbor, slotted collar, and nut shall be supplied fully assembled. The four setscrews shall be fitted to a plate secured to an inside surface of the carrying case.

Service data. The contractor shall enclose, with each unit, two copies of each of one or more booklets which contain complete instructions for installation, operation, maintenance, and a list of component parts. Wiring diagrams shall be included for electrical items or components. Each part of the unit other than common fasteners, etc., shall be identified in the service data and parts lists by the manufacturer's part number.

Identification marking. Each autopsy bone cutting saw shall be identified with a name plate located on the power handle (motor housing) bearing the name or registered trademark of the manufacturer, manufacturer's address, type or model number, serial number, voltage, full load current and cycles per second to be used for its operation. The name plate shall further bear the following warnings: "NOT APPROVED FOR HAZARDOUS LOCATIONS", and "DO NOT AUTOCLAVE OR IMMERSE IN LIQUID". The above specified warnings shall be in clear, large and legible characters. As an alternate to the nameplate warning, the following is acceptable on a separate nameplate "DANGER-EXPLOSION HAZARD. DO NOT USE IN THE PRESENCE OF FLAMMABLE ANESTHETICS".

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Workmanship. The autopsy saw shall be free from defects which detract from its appearance or impair its serviceability.

Unit. Each (EA). One autopsy saw, as specified, constitutes one unit.

QUALITY ASSURANCE PROVISIONS.

Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government upon request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

Inspection. Inspection, as used herein, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item

Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, with and AQL of 1.0 (percent defective) and an inspection level of II. The unit of product for examination purposes shall be one autopsy saw.

Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, with an AQL of 0 (percent defective) and an inspection level of S-3. The unit of product for test purposes shall be one autopsy saw.

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Tests. Tests shall be conducted to determine compliance with specification requirements. Where feasible, the same sample shall be used for the determination of two or more text characteristics.

Performance test. Sample bone saws shall be tested for the requirements specified herein and as listed on identification plate:

Amperes at full load and no load on 115 volts, 60 cycles, alternating current.

Oscill Oscillations per minute at no load and full

Horsepower output.

Breakdown time (in hours) when saw is operated under full load current.

Temperature rise of motor housing, under full load current, during 5 minutes operating period.

Motor Test. Each sample autopsy saw shall be tested to determine compliance with Underwriters' Laboratories, Inc. Standard UL 544 requirements for continuous rated universal motors for electric tools.

General performance tests. Before bone saws have been tested as specified herein, the bone saw motors shall be checked for switch operation, excessive noise and vibration.

Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producers' own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

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Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerance specified using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this document are met.

If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable.

The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements.

Federal Food, Drug, and Cosmetic Act. If the product covered by this document has been determined by the U.S. Food and Drug Administration to be under its jurisdiction , the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of the Federal Food, Drug, and Cosmetic Act, as amended, and regulations promulgated thereunder. In addition, the offeror/contractor shall comply, and be responsible for compliance by its subcontractors/suppliers, with the requirements of all other applicable Federal, State, and local statutes, ordinances, and regulations.

Recovered material. The offeror/contractor is encouraged to use recovered material in accordance with Federal Acquisition Regulation Subpart 23.4 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. Unless otherwise specified, preservation, packaging, and packing shall be to a degree of protection to preclude damage to containers and/or contents thereof under normal shipping conditions, handling, etc., involving shipment from the supply source to the receiving activity, plus reshipment from receiving activity, and shall conform to applicable carrier's rules and regulations. Intermediate and exterior package quantities and labeling and marking shall be as specified in the contract and/or order. Ordering data (Intermediate/exterior package quantities and labeling and marking must be specified in the contract and/or order).

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NOTE 1: The following National Stock Numbers are covered by this document:

NSN	ITEM IDENTIFICATION
6515-00-299-8717	SAW, BONE, ELECTRIC, OSCILLATING, AUTOPSY , 115 VOLTS, AC-DC.
6515-01-172-7676	SAW, BONE, ELECTRIC, OSCILLATING, AUTOPSY, 120/230 VOLTS, 50/60 HZ AC-DC.

NOTE 2: The National Stock Number for the blade specified herein is NSN 6515-00-299-8718, BLADE, BONE, SAW, ELECTRIC, OSCILLATING, AUTOPSY.

MILITARY INTERESTS:

Custodians:

Army-MD
Navy-MS
Air Force-03

PREPARING ACTIVITY:

DoD-MB

Agent:

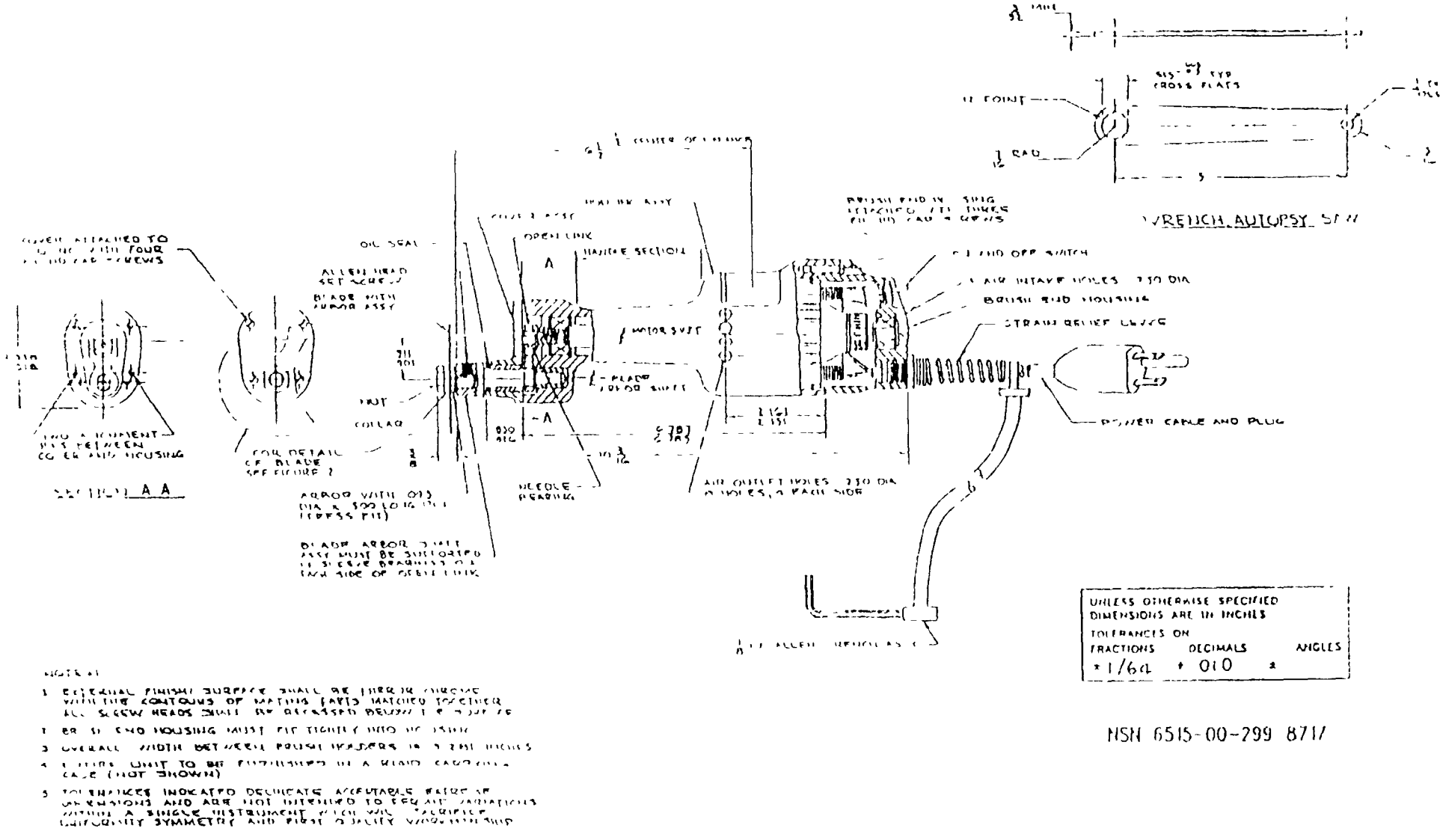
DLA-DM

CIVIL AGENCY COORDINATING ACTIVITIES:

VA-OSS
USPHS
FDA-MPQAS

Project No. 6515-5491

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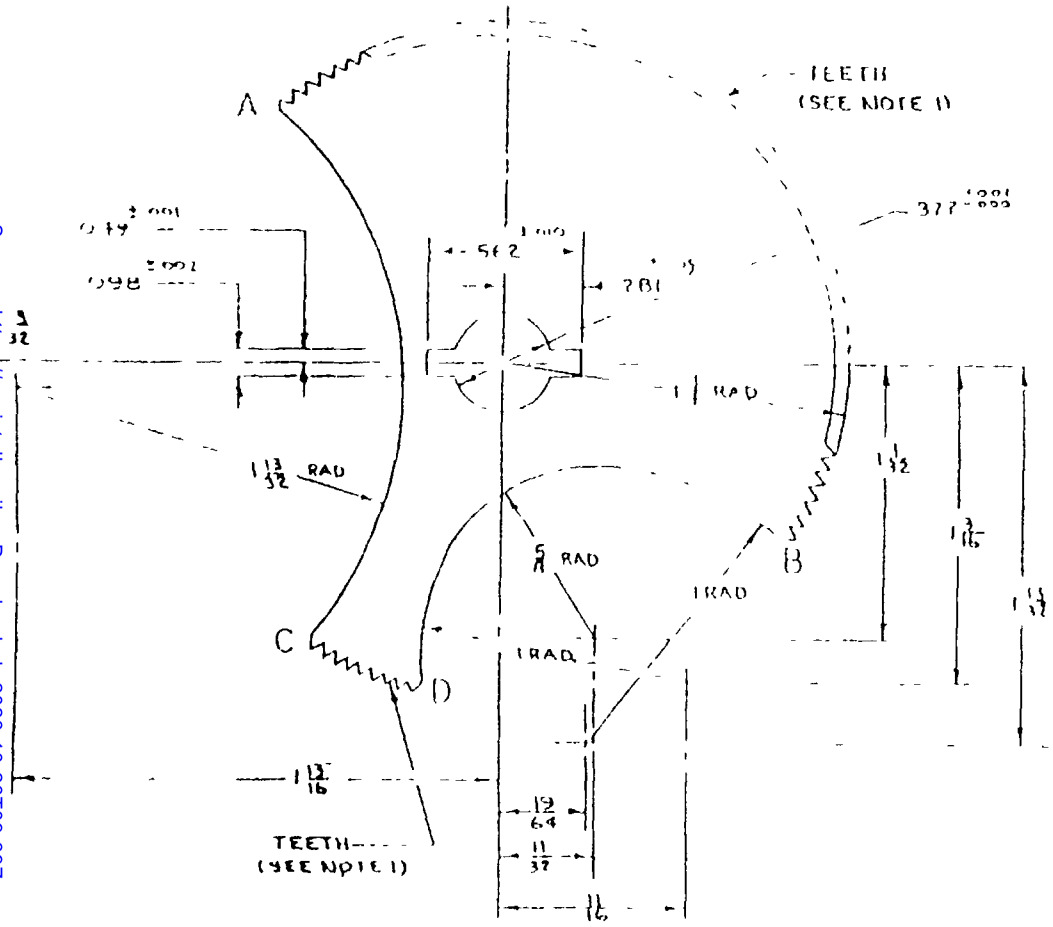
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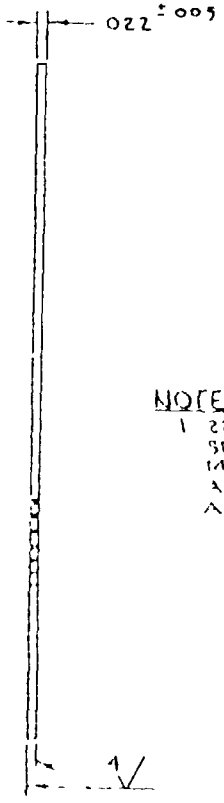
1. EXTERNAL FINISH SURFACE SHALL BE FREE OF SHARP EDGES WITH THE CONTOURS OF MATING PARTS MATCHED TOGETHER ALL SCREW HEADS SHALL BE RECESSED BELOW THE SURFACE
2. END HOUSING MUST FIT TIGHTLY INTO HOUSING
3. OVERALL WIDTH BETWEEN BRUSH HOLDERS IS 2.781 INCHES
4. LISTED UNIT TO BE FURNISHED IN A HARD CARRYING CASE (NOT SHOWN)
5. TOLERANCES INDICATED DECIMALS ACCEPTABLE RANGE OF DIMENSIONS AND ARE NOT INTENDED TO PERMIT VARIATIONS WITHIN A SINGLE INSTRUMENT WHICH WILL JEOPARDIZE CONCENTRICITY SYMMETRY AND FIRST QUALITY WORKMANSHIP

NSN 6515-00-799 8717

FIGURE 1 SAW, BONE CUTTING, AUTOPSY, STRYKER

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NOTE

1 22-60° "V" TYPE TEETH PER INCH SET .005 ± .002 EACH SIDE. MUST HAVE FULL TEETH AT EACH END OF ARCS A-B & C D

UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES		
TOLERANCES ON FRACTIONS	DECIMALS	ANGLES
± 1/32	-	=

NSN 6515-00-299-8718

FIGURE 2 BLADE, BONE SAW, ELECTRIC, AUTOPSY

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER: A-A-54841	2. DOCUMENT DATE (YYMMDD): 31 March 1993

3. DOCUMENT TITLE: SAW, BONE, ELECTRIC, OSCILLATING, AUTOPSY

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 — Previous editions are obsolete. 96-290

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